

PROJECT GUIDELINES

Batch Project

Team Size: 5 Members

Total Groups: 6

Duration: 10 days

Objective of the Project

This is not just a model-building exercise.

You are expected to demonstrate:

- Structured problem solving
- Engineering discipline
- Team collaboration & role structure
- Business thinking
- Technical depth
- Professional presentation

This project simulates how enterprise data science teams operate.

Timeline Structure

Presentation: Saturday 11 April 2026

Deliverables: Discovery + Design + Baseline + Implementation

Focus on:

- Understanding the business problem
- Data exploration
- Identifying challenges
- Designing architecture (HLD + LLD)
- Building a baseline solution

You must present:

- Your understanding of the problem
- Data insights
- Identified risks & challenges, edge cases
- High-Level Design (HLD)
- Low-Level Design (LLD draft)
- Results

Your Presentation will be Evaluated as per below metrics

1. Research, Scoping & Problem Framing (Points: 4)
2. Business Relevance & User Journey (Points: 4)
3. Technical Clarity & Architecture (Points: 4)
4. Demo Completeness (Points: 4)
5. Repo / Evidence Trail (Points: 4)

Note: The scores will be provided individually and not as a group. Therefore in the presentation each individual member must provide proof and clarity of their own contribution. Members who do not participate in presenting their contribution will score a zero (0).

The first slide of the presentation must clarify the team structure and roles for the purpose of evaluation.

Final Project Submission Day 18th April 2026

Refinement (based on feedback) + Modeling + Business Translation

The project must be improved or fine tuned based on feedback received. Focus on:

- Advanced feature engineering
- Model comparison
- Validation strategy
- Performance optimization
- Error analysis
- Business recommendations

Final Submission: 18th April 2026

The Final Project will be Evaluated as per below metrics

1. Research, Problem Selection & Scope (Point: 4)
2. Technical Implementation & Innovation (Point: 4)
3. Requirement Coverage & Robustness (Point: 4)
4. Engineering / Repo Quality (Point: 4)

5. Feedback Incorporation (Point: 4)

As with presentation, each member will be scored individually based on their contribution.

Team Structure (5 Members)

Do NOT divide work sequentially.

Divide by vertical ownership.

Suggested Role Structure

1 Problem & Business Lead

- Define business objective clearly
- Define success metric
- Translate results into business insights
- Lead final storytelling

2 Data Engineering Lead

- Data cleaning & preprocessing
- Integration & transformations
- Missing value handling strategy

3 Feature Engineering Lead

- Create meaningful features
- Document hypothesis behind each feature
- Manage text/categorical encoding (if applicable)

4 Modeling & Evaluation Lead

- Build baseline
- Compare multiple models
- Design validation strategy
- Handle imbalance (if applicable)

5 Engineering & Documentation Lead

- Repository hygiene
- Modular code structure
- Maintain README
- Track GitHub issues
- Manage AI usage documentation

6 UI & Demonstration Lead

- Design and develop a simple interface (e.g., Streamlit or other framework)
- Connect trained model to the interface for live/demo predictions
- Ensure clean user flow and clear output explanation
- Handle deployment setup (local/cloud) for presentation
- Prepare demo script to showcase business use-case clearly

Every member must:

- Commit code
- Contribute intellectually

Repository Standards (Mandatory)

Your GitHub repository must contain:

```
project-root/
|
├── data/
│   ├── raw/
│   └── processed/
├── .py files/
├── src/
├── docs/
│   ├── HLD.md
│   └── LLD.md
├── README.md
├── requirements.txt
└── PROMPTS.md
```

Rules:

- No last-day bulk commits
- Clear commit messages
- Proper .gitignore

- Modular code only.
- Code plagiarism will be punished.

Commit history is part of evaluation.

Design Requirements (Before Heavy Modeling)

HLD – High Level Design

Must include:

- Business problem explanation
- Stakeholder perspective
- Data flow architecture
- Chosen evaluation metric & why
- Assumptions & risks

LLD – Low Level Design

Must include:

- Feature list
- Preprocessing steps
- Missing value strategy
- Model hypotheses
- Validation strategy

Design first. Code later.

Modeling Expectations

Every group must:

1 Build a Baseline Model

- Simple model

- Proper evaluation
- Honest metrics

2 Compare Multiple Models

Minimum 3 approaches.

3 Use Correct Evaluation Metric

Accuracy alone is rarely sufficient.

Choose metrics aligned with the business problem.

4 Perform Error Analysis

- Where does the model fail?
- Pattern in misclassifications?
- Edge cases?

5 Threshold / Decision Strategy (if classification)

Explain how business would use predictions.

Final Project Submission Structure

1. Business Context
2. Problem Framing
3. Data Story
4. Feature Strategy
5. Modeling Approach
6. Model Comparison
7. Evaluation Metrics
8. Error Analysis
9. Business Recommendations
10. Future Improvements

Present professionally.

AI Usage Policy

AI tools are allowed.

But:

- Usage of AI should not be more than 20-30%, Code should not be copied from the AI.
- You must understand every line of code.
- Document how AI helped.
- Do not blindly copy output.
- You must be able to defend technical decisions live.

Common Mistakes

- One person doing everything
- Ignoring design documents
- Reporting only accuracy
- No model comparison
- No business interpretation
- Poor Git discipline
- Weak storytelling
- Unbalanced participation

What Impresses

- Clear thinking
- Strong feature hypotheses
- Structured experimentation
- Clean architecture
- Honest discussion of limitations
- Strong collaboration
- Consultant-level presentation

“Your project will not be evaluated only on what you build, but on how professionally, collaboratively, and thoughtfully you build it.”

